

DHARMA TEJA GURRAM

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SUMMARY

ASIC/SoC Design Verification Engineer with CPU-subsystem and IP-level verification experience across AMBA AHB/APB peripherals, ARM Cortex-M0+ based subsystems, and mixed-signal (AMS) blocks. Builds boot/bring-up and trace-based reference-model verification infrastructure, drives UVM-based test development to functional and code coverage closure, and develops EDA scripting and automation tooling that speeds up verification cycles.

EDUCATION

Indian Institute of Technology (IIT) Mandi

Bachelor of Technology, Electrical Engineering (Honors) — CGPA: 8.68/10

Mandi, India

2020 – 2024

Coursework: Computer Organization, Reconfigurable Computing, Computing & Data Science, Digital System Design, Digital MOS LSI Circuits, Data Science, Estimation and Detection, Probability & Statistics

TECHNICAL SKILLS

- **Verification Methodology:** UVM (driver, monitor, scoreboard, sequencer), SystemVerilog Assertions, functional & code coverage closure, Tarmac trace-based reference-model checking
- **Protocols & Subsystems:** AMBA AHB/APB, ARM Cortex-M0+ based CPU subsystem verification (boot, ISR/vector table, interrupts), memory-mapped register (MMR) verification
- **EDA & Tools:** Cadence Xcelium, SimVision, JasperGold (Lint/CDC/Formal), vManager, Virtuoso, Makeflow (ARMCC, ARMClang, ARMLink)
- **Programming & Scripting:** Python, C/C++, Tcl, Bash, Csh
- **Design & HDL:** SystemVerilog, Verilog, VHDL, Verilog-A
- **Version Control & CI/CD:** Perforce (Helix Core), Design Sync, Git, Bitbucket, Jira, Confluence
- **Automation & Tooling:** Python-based EDA flow automation, LLM-agent-driven regression triage and debug tooling, React/Node dashboards, AWS (EC2, S3, RDS, Lambda), Docker

PROFESSIONAL EXPERIENCE

Design Verification Engineer | Texas Instruments, Bengaluru, India

Jul 2024 – Present

- **CPU Subsystem Verification (new project bring-up):** Owned DV for the ARM Cortex-M0+ based CPU subsystem; built the C boot flow and image build chain from scratch (ARM CC/Link, ELF-to-hex, RAM/ROM loading), authored the ISR vector table and boot bring-up setup, and integrated Tarmac trace-based reference-model checking against an instruction-set simulator for instruction-level cross-validation.
- **Subsystem & IP Test Development:** Developed and owned interrupt, memory-mapped register (MMR), and subsystem-level functional tests within the team's UVM verification environment (test-class extension of `base_test`, `run_phase`-based sequencing), driving closure against functional and code coverage metrics; built a test-template generation tool and a recompile-only-the-C-build flow that cut boot-code iteration time without re-elaborating the RTL.
- **IP-Level & Mixed-Signal DV:** Owned DV for ADC (SAR ADC controller, calibration control logic, analog/AMS behavioral models), OTP (memory array and controller), and DPWM/Filter (functional) — testcase/checker/assertion development, AMS model debug, and sanity/regression closure in Xcelium/vManager.
- **Verification Automation Infrastructure:** Architected an LLM-powered, state-aware agentic automation stack that turns manual DV workflows into autonomous CI/CD-style pipelines — maximizing off-hours compute utilization and shrinking project timelines.
 - **Autonomous Regression Lifecycle Manager** 🌀 — hands-off agent handling workspace syncs, regression triage, local fixes and debug reporting; reclaims idle off-hours compute, saving ~2 person-months per ~1.5-year cycle and freeing lead-engineer bandwidth.
 - **Spec-Aware Semantic-Temporal Debug Agent** 🌀 — MCP-based closed-loop AI debug agent enabling natural-language querying of simulation data via a custom semantic query adapter (`simvisdbutil/VCD`) to isolate relevant signal windows and root-cause failures, cutting debug turnaround by 6x.
 - **Release-Triggered Quality Gate** 🌀 — listens for release emails, auto-launches per-IP Lint/CDC checks, monitors LSF job health, and posts violation summaries to Confluence.
- Drove BU-wide adoption of AI-assisted engineering (Claude Code, custom MCP integrations) across design/verification teams.

Digital Design Intern | Texas Instruments, Bengaluru, India

Jan 2023 – Jul 2023

- UCD3138 Verification — implemented interleaved (2/3-phase) peak-current-mode control from system-level requirements and verified it end-to-end in Xcelium/SimVision: Verilog/AMS debug and full pad-to-pad data-path verification, bringing the design up to a working SoC condition with the system engineer.
- Synthesized IP blocks (I²C, PMBus, Timer) for area/power/timing reports and netlists; authored an implementation spec from a communication-interface IP analysis.

PROJECTS

AMBA AXI4-Lite-to-APB Bridge 🌀 — UVM Verification Environment Built From Scratch

SystemVerilog · UVM · Verilator/Icarus

- Designed and verified an AXI4-Lite-to-APB bridge (AMBA) and an original register-mapped peripheral in SystemVerilog — from-scratch bridge FSM, UVM-methodology environment (driver/sequencer/monitor/scoreboard, TLM-style analysis ports), and a behavioral reference-model scoreboard checking functional correctness rather than data echo; closed functional coverage (94% on randomized testing) and wrote SystemVerilog Assertions for protocol legality, with all 8 tests passing at 0 errors via `make sim` from a clean rebuild.
- Root-caused two distinct bugs while bringing the environment up from scratch: a blocking/non-blocking assignment timing race in the test-bench, and an actual Verilator simulator bug reproduced across two Verilator versions, which I bisected to a minimal repro and worked around.

MACTrace 🌀 — C++ Functional & Performance Model of a Configurable CNN Accelerator

C++17 · CMake · PyTorch · Python

- Built a C++ functional model of a configurable CNN accelerator (MAC array, accumulator, SRAM buffer classes) mapping convolution and fully-connected layers onto tiled GEMM execution, validated against a PyTorch-trained INT8-quantized MNIST model.
- Instrumented the accelerator with performance counters (MAC utilization, cycle estimates, buffer traffic) to compare hardware configurations, measuring utilization differences between convolution and fully-connected layers on the same workload.

NeuronEase 🌀 — Memristor-Based Hardware Accelerator for Neural Networks

Cadence Virtuoso · Verilog-A · Verilog · TensorFlow

- Final-year project: designed/simulated a memristor-crossbar accelerator (Stanford ReRAM, Cadence Virtuoso, Verilog-A) for analog multiply-accumulate — weights as crossbar conductances, DAC/ADC I/O; validated via a SPICE-level 2×2 matrix-multiplication demo.
- Trained a compact CNN in TensorFlow (3×3 , 12-kernel conv + max-pool + dense, 20,410 INT8 params) and mapped it onto a 9×8 memristor crossbar (row/col decoders, DAC/ADC, conv controller); verified against MNIST at ~12 cycles/pixel, and separately fabricated/characterized Cu/TiO₂/Pt memristive cells (VLSI Fabrication Practicum), measuring I-V resistive switching (I_{on}/I_{off} up to 10^4); also built a related FCNN accelerator on a Zybo Z7 FPGA (Verilog, TensorFlow-trained).